

Torque comparison of the 60s 3 phase and 5 phase SynRM models

Issue #45: zero third harmonic current test

September 25, 2026

1 Short summary

This report summarizes the tests performed while comparing `synrm_3f_60s` and `synrm_5f_60s` for issue #45. The purpose of the test is to compare the two 60s models using the same rotor and the same fundamental dq current, while setting the third harmonic current of the five phase model to zero

$$i_{d3} = i_{q3} = 0$$

The comparison was carried out in several steps. First, the effect of the different turns per coil, N_c , was tested. Second, the transient simulation was repeated over one complete electrical period. Third, a fixed current magnitude angle sweep was performed. Finally, the same angle sweep was repeated after changing the 5f `InitPos` from -45° to -42° .

2 Test setup

The two models use the same 60s stator geometry and accept the same rotor barrier geometry. Their slots per pole per phase are different

$$q_{3f} = 5, \quad q_{5f} = 3$$

For each comparison, the same rotor geometry was used for both machines. The same fundamental dq current (i_d, i_q) was applied to both models, while the third harmonic current of the five phase machine was set to zero, $i_{d3} = i_{q3} = 0$.

The test implementation is in `tests/test_synrm_60s_torque_i3_zero.py`.

For the angle sweep, the current magnitude was kept fixed at

$$I = \sqrt{i_d^2 + i_q^2} = 3\sqrt{2}$$

with

$$i_d = I \cos \beta, \quad i_q = I \sin \beta$$

and β was varied from 35° to 85° in 5° steps.

3 Sequence of checks

3.1 Original turns per coil

The first comparison used the turns per coil inherited by the two models

$$N_{c,3f} = 68, \quad N_{c,5f} = 113$$

With the same rotor and $i_{d3} = i_{q3} = 0$ on the five phase side, the following values were obtained:

(i_d, i_q)	N_c (3f / 5f)	T_{3f} [Nm]	T_{5f} [Nm]	Relative difference
(1.5, 1.5)	68/113	8.28	15.91	92%
(3, 3)	68/113	16.33	28.25	73%

3.2 Equal turns per coil

The test was then repeated with

$$N_{c,3f} = N_{c,5f} = 113$$

Two rotor seeds were checked. The reported values were:

(i_d, i_q)	Seed	T_{3f} [Nm]	T_{5f} [Nm]	Relative difference
(1.5, 1.5)	42	13.82	15.91	15.1%
(1.5, 1.5)	43	14.24	16.20	13.8%
(3, 3)	42	24.86	28.25	13.6%
(3, 3)	43	26.24	30.21	15.1%

3.3 Full period transient check

Both machines were also tested with

$$N_{\text{per}} = 1, \quad \text{PointPer} = 301$$

The relative torque differences were 13.03% for seed 42 and 13.97% for seed 43.

4 Current angle sweep

The current magnitude was kept constant at

$$I = 3\sqrt{2}$$

with

$$i_d = I \cos \beta, \quad i_q = I \sin \beta$$

For the five phase model, the third harmonic current was set to zero

$$i_{d3} = i_{q3} = 0$$

4.1 Five phase `InitPos` = -45°

The first current angle sweep used the original five phase setting, `InitPos` = -45° .

The results are listed in Table 1.

Table 1: Torque results with five phase `InitPos` = -45° .

β [deg]	T_{3f} [Nm]	T_{5f} [Nm]	Difference [%]
35	19.718	23.574	19.56
40	22.418	26.120	16.52
45	24.861	28.254	13.64
50	27.034	29.880	10.53
55	28.804	31.347	8.83
60	30.227	32.149	6.36
65	31.339	32.529	3.80
70	31.468	30.740	-2.31
75	30.124	25.169	-16.45
80	25.424	13.161	-48.23
85	14.962	-3.682	-124.61

Among the sampled current angles, the largest three phase torque was 31.47 Nm at $\beta = 70^\circ$, while the largest five phase torque was 32.53 Nm at $\beta = 65^\circ$.

4.2 Five phase InitPos = -42°

The value -42° was tested based on the change in the winding reference between the original 40s and the new 60s five phase models. With the change from $q = 2$ to $q = 3$, the calculated phase A fundamental winding axis shifts by approximately 6° electrical. For two pole pairs, this corresponds to 3° mechanical

$$6^\circ / 2 = 3^\circ$$

The original 40s model used InitPos = -45° , so the corresponding compensated value tested for the 60s model was

$$-45^\circ + 3^\circ = -42^\circ$$

The same current angle sweep was then repeated with InitPos = -42° for the five phase model.

The results are listed in Table 2. Among the sampled current angles, the largest torque occurred at $\beta = 70^\circ$ for both models. The corresponding torque values were 31.47 Nm for the three phase model and 32.31 Nm for the five phase model.

Table 2: Torque results with five phase InitPos = -42° .

β [deg]	T_{3f} [Nm]	T_{5f} [Nm]	Difference [%]
35	19.718	19.970	1.28
40	22.418	22.698	1.25
45	24.861	25.231	1.48
50	27.034	27.613	2.14
55	28.804	29.376	1.99
60	30.227	30.807	1.92
65	31.339	31.975	2.03
70	31.468	32.315	2.69
75	30.124	31.049	3.07
80	25.424	26.539	4.38
85	14.962	15.991	6.87

5 Observed changes

The sequence of tests can be summarized as follows:

Check	Main change	Recorded observation
Original comparison	$N_c = 68$ (3f) $N_c = 113$ (5f)	73–92% torque difference for the two tested current pairs
Equal N_c comparison	Both $N_c = 113$	13.6–15.1% difference across the tested currents and seeds
Full period check	$N_{per} = 1$ PointPer = 301	13.03% and 13.97% for seeds 42 and 43
Angle sweep, original reference	$N_c = 113$ 5f InitPos = -45°	Maximum sampled torque at 70° for 3f and 65° for 5f
Angle sweep, modified reference	$N_c = 113$ 5f InitPos = -42°	The torque values at different angles are close to each other

6 Confusions

Two points remain unclear:

- How should N_c be chosen for the 3f and 5f comparison? Should both models use the same N_c , or should each winding keep its own value?
- How should `Ini tPos` be chosen for the 5f 60s model?